On my honor I have neither given nor received any unauthorized aid for in appropriate assistance for all sessions of this exam. The work done on this exam is totally my own I understand that by school code a violation of these principles will lead to a zero grade and is subject to harsh dissiplinary issues

Question 1 a) First by I will use the 2nd elimination matrix

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$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & \gamma & 1-8\alpha \\ 2+m & \beta+m\alpha & 3-m\gamma & \alpha^2+m\gamma \\ 0 & 14-\alpha & \gamma-\alpha & 2-\alpha \end{bmatrix}$$

from this elimination "$3\alpha-8$" and "$2+m$" must be zero

$3\alpha-8=0 \rightarrow \boxed{\alpha=\frac{8}{3}}$

$2+m=0 \rightarrow \boxed{m=-2}$

1st elimination matrix

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & (-49/3) & \gamma & (-56/3) \\ 0 & \beta-(16/3) & 3+2\gamma & \frac{64}{9}-2\gamma \\ 0 & 34/3 & \gamma-(8/3) & \frac{-2}{3} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & \gamma & -56/3 \\ 0 & \beta+11 & 3+\gamma & \frac{232}{9}-2\gamma \\ 0 & \frac{34-49k}{3} & (k+1)\gamma-\frac{8}{3} & \frac{-2}{3}-k\cdot\frac{56}{3} \end{bmatrix}$$

$\beta+11=0$

$\boxed{\beta=-11}$

$\frac{34-49k}{9}=0$

$\boxed{k=\frac{34}{49}}$ $\boxed{k=0,694}$

3rd elimination

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & +\gamma & -56/3 \\ 0 & 0 & 3+\gamma & \frac{232}{9}-2\gamma \\ 0 & 0 & (1,694)\gamma-\frac{8}{3} & (13,62) \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & \gamma & -56/3 \\ 0 & 0 & 3+\gamma & (25,78)-2\gamma \\ 0 & 0 & (1,694)\cdot\gamma-\frac{25}{6} & 0,73-\gamma \end{bmatrix}$$

$1,194\cdot\gamma-\frac{25}{6}=0$

$\gamma=\frac{25}{6\cdot(1,194)}$

$\boxed{\gamma=3490}$

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b)

$$\begin{bmatrix} 1 & 2,667 & -3,490 & 2,667 \\ 0 & -6,33 & 3,490 & -18,67 \\ 0 & 0 & 6,490 & 18,72 \\ 0 & 0 & 0 & 2,760 \end{bmatrix}$$

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Question 2 ($X = SY$)

a) "k" must be number that k-dimantion would have include the x-axis and a y-axsis.

Also this shape must have a 3D area inside because "The intossetion" must have a volume which is a 3D consept. This can be achived If the shape is MINIMUM 3D.

And "$X = SY$" can only be 3D when this system is build in a 4D envirement.

So $\boxed{K_{min} = 4}$

b) For this expression to be correct in rref(A) there must be 3 Free colums. $ax + by + cz + dt = 0$

$$\begin{bmatrix} 1 & -5 & 0 & 0 \\ a & b & c & d \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \left[\begin{array}{cccc|c} 1 & -5 & 0 & 0 & 0 \\ 0 & b+5a & c & d & 0 \end{array}\right]$$

→ As lony as "$b+5a$" is equal to Zero we will have 3 Free colums $b+5a = 0$ $b = -5a \Rightarrow ax - 5ay + cz + dt = 0$

$(a, c, d \in \mathbb{R})$

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C) $x - 5y = 0$

$x + 4y - z = 6$

4D $\rightarrow \underbrace{\begin{bmatrix} 1 & -5 & 0 & 0 \\ 1 & 4 & -1 & 0 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$

$\text{rref}(A) \Rightarrow \begin{bmatrix} 1 & 0 & -5/9 & 0 \\ 0 & 1 & -1/9 & 0 \\ P & P & F & F \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 10/3 \\ 2/3 \end{bmatrix}$

$x + \frac{-5}{9}z = \frac{10}{3} \Rightarrow x = \frac{10}{3} + \frac{5}{9}z$

$y + \frac{-1}{9}z = \frac{2}{3} \Rightarrow y = \frac{2}{3} + \frac{1}{9}z$

$z = \alpha$ $t = \beta$

Possible vectors $\Rightarrow \begin{bmatrix} \frac{10}{3} \\ \frac{2}{3} \\ 0 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 5/9 \\ 1/9 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (\alpha, \beta \in \mathbb{R})$

This is NOT a vector space because it is not multiplyable (ie) $f(5x) \neq 5f(x)$

The intersection has 2 dimention because the possible vectors only have 2 Free colums, In a 4D ($\mathbb{R}^4$) space because "$x = 5y$" must be in an at least 4D space because of that we have 4 colums in total

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Question 3)

$H = I_n - 2UU^T$ (in $\mathbb{R}^n$) (U is a unit vector

a)

$H^2 = H \cdot H = (I_n - 2UU^T) \cdot (I_n - 2UU^T) = I_n \cdot I_n - 2UU^T \cdot I_n - I_n \cdot 2U \cdot U^T + 4UU^T UU^T$

$= I_n - 2UU^T - 2UU^T + 4UU^T UU^T$

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Question 4) a)

The transfered amount is $c \cdot (V(t) - W(t))$ $(|c| \leq 1)$. The total amount of water must stay the same which is 50 liters

$$V_{k+1} = V_k + c(V_k - W_k) = V_k(c+1) - c \cdot W_k$$

$$W_{k+1} = W_k - c(V_k - W_k) = W_k(c+1) - c V_k$$

$$\begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} (c+1) & -c \\ -c & (c+1) \end{bmatrix} \begin{bmatrix} V_k \\ W_k \end{bmatrix}$$

the change is

$$\frac{dV(t)}{dt} = +c(V_k - W_k) = cV_k - cW_k$$

$$\frac{dW(t)}{dt} = -c(V_k - W_k) = -cV_k + cW_k$$

$$\frac{d}{dt}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} c & -c \\ -c & c \end{bmatrix} \begin{bmatrix} V_k \\ W_k \end{bmatrix}$$

If "c=-1" then

$$\frac{d}{dt}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} V \\ W \end{bmatrix}$$

b) $V_1 = V_0(-1+1) - (-1)W_0$

$V_1 = W_0 = 40$ liters

$W_1 = V_0 = 10$ liter

$V_{k+1} = W_k$

$W_{k+1} = V_k$

c) There wouldn't be a stead, state because the water levels will just swich with eachotan to infinite

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